



Home Automation And Security Using IoT Devices

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Home sweet home—there's nothing like it! Our homes can be made smarter using the Internet of Things (IoT)—a system in which computing devices are connected over a network using unique identifiers (UIDs), and which can transfer data without the need for any human-to-human or human-to-computer interactions. These IoT devices can be used to automate our homes also

IoT is a system in which computing devices are connected over a network using unique identifiers (UIDs) and can transfer data without the need of any human-to-human or human-to-computer interaction. Such computing devices are called IoT devices and can be used to automate our home as well.

The first home automation technology was developed in 1975 using a network technology called X10, which is a communication protocol that was used for home automation devices back then. It used electric power transmission wiring for signaling and con-

trolling various electronic devices. The signals used radio frequency as digital data to control any digital electronic device installed in a house.

Technology has changed now; with the help of sophisticated hardware and free open source software (FOSS), home automation can make a dream house come to reality.

Basic setup for home automation

The setup for home automation using IoT devices requires a router connected to the Internet and a controller/hub that is connected to the IoT devices or sensors.

There are four primary operating standards for home automation technology—Wi-Fi, Z-Wave, Zigbee, and Bluetooth Low Energy (BLE). Wi-Fi is the most commonly used standard where IoT devices are connected to a centralised controller/hub (either wired or wireless), which is connected to the router through an Ethernet link.

The IoT devices can connect to a cloud network on the Internet using the controller/hub and the router. They can be controlled through the cloud network using a FOSS or a mobile app. The mobile app can be used to send and receive data through the cloud network to control and monitor the IoT devices installed in a house.

The above architecture gives the convenience to control and automate various IoT devices from a distant location with the help of a smartphone and the Internet. The IoT devices will be able to send data like room temperature, supply voltage, energy consumption, live video stream, etc to the smartphone app through the Internet and enable remote monitoring of our homes.

IoT devices for securing your home

There are various IoT devices and sensors that can be used for home automation and security. The basic ones are:

1. Wall switches. IoT-based wall switches can be mounted on the wall of a house like regular wall switches, but can be controlled remotely. Such IoT switches can also be wireless and can be connected to an IoT hub or router wirelessly.

2. Voltage sensors. IoT voltage sensors can monitor the supply voltage of the house and one can even monitor the same remotely using the Internet.

3. Energy monitors. IoT energy monitors are digital energy meters that can monitor the power consumption of the house, also helping one monitor the overall power consumption remotely.

4. Thermostats. IoT thermostats can be used to monitor the temperature